

Computational Geometry (2025)

Theory Questions

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Question 1 (Rectangle Stabbing Counting (★★)): Use the segment tree paradigm to solve the following problem: given a set S of n axis-aligned rectangles in two dimensions, build a data structure such that given a query point p , one can *count* the rectangles that contain q efficiently. Your data structure should use $O(n \log n)$ space and should be able to answer queries in $O(\log^2 n)$ time.

Question 2 (Geodesic Distances in Polygons with Holes (★★)): Let P be a polygon and let q_1, q_2 be two points inside P (or on its boundary). The *geodesic distance* between q_1 and q_2 is the length of the shortest path that connects q_1 to q_2 and it does not go outside P (i.e., it does not cross onto the outside region or into any of the holes inside P). For a point q inside P , the point that is *farthest* away from q is a point q' that maximises the distance between q and q' (and q' is either on the boundary of P or inside P). See Figure 1.

Give an example of a polygon P with holes and a point q inside it such that the point q' that is farthest away from q is **not** on the boundary P and it lies entirely **inside** P .

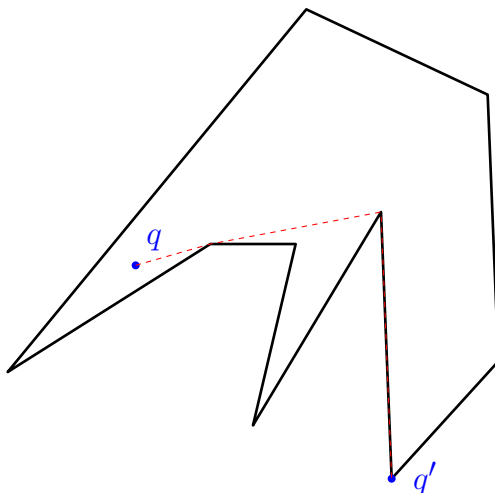


Figure 1: The point q' is the point that is farthest away from q inside the polygon. If the polygon does not have holes, then q' will always be a boundary point.

Question 3 (area computation (★★★), (★★★★) for the divide-and-conquer solution): Let R be a set of n axis-aligned rectangles (i.e., rectangles with sides parallel to X - and Y -axes). Show that it is possible to compute the area of the union of the rectangles in $O(n \log n)$ time. See Figure 2.

Hint: I know both a divide-and-conquer solution as well as a sweep line solution. You will get bonus points if you can figure out the divide-and-conquer approach!

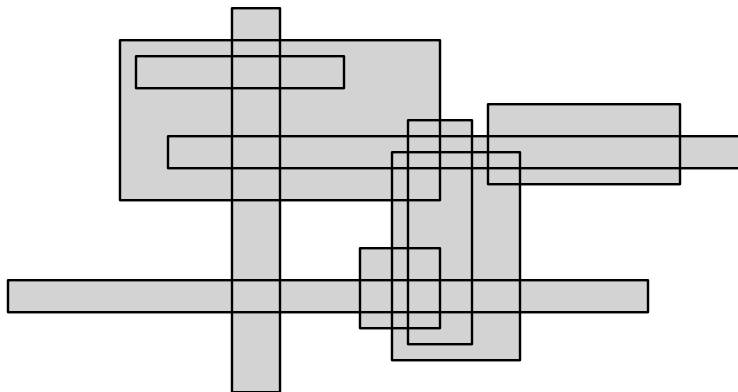


Figure 2: The goal is to compute the area of the grey region.

Question 4 (Halfspace range max queries (*)):** Consider an input set P of n weighted points in the plane, meaning, each point $p_i \in P$ is associated with a real-valued weight $w(p_i)$. The points and their weights are given as input. Prove that it is possible to build a data structure of linear size such that given a query line ℓ , the point with the largest weight below ℓ can be found in $O(\log n)$ time.

Question 5 (Minkowski Sum (*)):** Consider the following statements. For each statement, decide if it is true or not. If true, then provide a proof otherwise give a counterexample. Here $P \oplus Q$ denote the Minkowski sum of two polygons.

- (a) If P is a simple polygon then $P \oplus \text{Conv}(P)$ is convex, where $\text{Conv}(P)$ is the convex hull of P . This part is a bit harder than the other parts.
- (b) If P is a simple polygon and C is a disk (circle and its interior) then $P \oplus C$ is convex if the diameter of C is at least as large as the diameter of P .
- (c) If P is a simple polygon then there exists a large enough disk such that $P \oplus C$ is convex.
- (d) For two convex polygons P and Q , the perimeter of $P \oplus Q$ is equal to the sum of the perimeters of P and Q .

General remarks. Have the duality transformation in mind. Some of the questions become much more manageable if you look at the dual problem. Also expected bounds are acceptable. Finally, if you are stuck on a problem, try to review the lectures and the topics covered in the class and perhaps you suddenly remember something that helps you solve the problem.

Difficulty Notation.

- (*) This is an “easy” question, meaning, the solution requires just one idea. It does not mean that figuring out the one idea is easy but for these questions you basically need a “light bulb to turn on” in your head. These questions have very short solutions; sometimes the solution can just be one line!
- (**) This is the next step. You need the initial idea, but even after having the initial idea, you will still have one or two more steps to go. These steps are typically much easier than getting the first initial idea but they still require some care.
- (***) This signifies that the problem requires a few somewhat complicated steps, or that the solution is longer than average.
- (****) This is a challenging problem. It should be solvable if you spend enough time on it.

(★★★★) These problems are very difficult to solve. This often means that it requires you to go beyond the ideas covered in the class and be innovative.

(★★★★★) So you want to stay up many nights in bed, thinking about this problem?